

Introduction to Statistics

Course Notes, Week 1

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Lecture 1: Introduction to Statistics

Statistics is the science of collecting, analyzing, and interpreting data. In this lecture, we'll explore the fundamental concepts.

Key Concepts

- **Population:** The entire group being studied
- **Sample:** A subset of the population
- **Parameter:** A numerical characteristic of a population
- **Statistic:** A numerical characteristic of a sample

Probability Basics

The probability of an event A is denoted as $P(A)$ and satisfies:

$$0 \leq P(A) \leq 1 \tag{1}$$

For two events A and B :

The **median** is the middle value when data is sorted.

Mean (Average) Definition 1.1

The **mean** of a dataset is the sum of all values divided by the number of values. It represents the central tendency of the data and is sensitive to outliers.

For a sample: $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$

Variance and Standard Deviation

The sample variance is:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \tag{2}$$

And the standard deviation is simply $s = \sqrt{s^2}$.

Lecture 2: Hypothesis Testing

The Framework

1. State null hypothesis H_0 and alternative H_A
2. Choose significance level α (typically 0.05)
3. Calculate test statistic
4. Make decision based on p-value

t-Test Example

For a one-sample t-test:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \quad [3]$$

where μ_0 is the hypothesized population mean.

t-Distribution

Theorem 2.1

Under the null hypothesis, if the population is normally distributed, the test statistic follows a t-distribution with $n - 1$ degrees of freedom.

Lecture 3: Conclusion

This covers the basics of statistical analysis. Next lecture will focus on regression analysis and correlation.

Using Colored Boxes

You can use the colored theorem boxes by inserting raw Typst code:

Central Limit Theorem

Definition 3.1

The distribution of sample means approximates a normal distribution as the sample size becomes larger, regardless of the population's distribution shape.

Law of Large Numbers

Theorem 3.1

As the number of trials increases, the sample mean converges to the expected value.

Sample Mean Calculation

Example 3.1

For the dataset $\{2, 4, 6, 8, 10\}$, the mean is:

$$\bar{x} = \frac{2 + 4 + 6 + 8 + 10}{5} = 6 \quad [4]$$

Important Note

Note 3.1

Always check your assumptions before applying statistical tests.

Warning

Attention 3.1

Sample size must be at least 30 for the Central Limit Theorem to apply reliably.

Available box types: - `#definition[]` - Olive colored (for definitions) - `#theorem[]` - Navy colored (for theorems) - `#proposition[]` - Maroon colored (for propositions) - `#example[]` - Purple colored (for examples) - `#note[]` - Blue colored (for

notes) - `#attention[]` - Red colored (for warnings/attention) - `#quote[]` - Black colored (for quotes) - `#hypothesis[]` - Orange colored (for hypotheses)

End of lecture notes